Hekai Wang

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Research Portfolio: hekaiwang.github.io

EDUCATION

University of Chinese Academy of Sciences (UCAS) — **Beijing / Shenzhen, China**

- M.S. in Computer Technology — **Sep 2024 – Jun 2027 (Expected)**
- Core Courses: Pattern Recognition and Machine Learning, Applied Matrix Theory, Cloud Computing Technology and Engineering, Algorithm Design and Analysis, Robotics and Intelligent Computing.

Hubei University of Technology — **Wuhan, China**

- B.E. in Mechanical Engineering — **Sep 2019 – Jun 2023**
- Core Courses: Signals and Systems, Measurement and Control Instruments, Control Engineering, Measurement and Control Circuits, Sensor Technology, Programmable Logic Controllers.

RESEARCH EXPERIENCE

Graduate Student Research Assistant — **2024 – Present**

Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences — **Shenzhen, China**

Supervised by Prof. Haifeng Wang and Associate Professor Yihang Zhou

- Investigated robust medical image segmentation across abdominal and brain MRI, focusing on anatomical priors, incomplete-modality learning, systematic modality-subset evaluation, and uncertainty analysis.
- Developed cross-modal medical imaging pipelines for MRI synthesis and biplanar X-ray-to-CT reconstruction, incorporating geometric alignment, generative modeling, image-quality assessment, and downstream segmentation validation.
- Conducted clinically oriented imaging research involving pediatric epilepsy outcome prediction and chest radiograph report review, including data processing, model validation, clinician-referenced evaluation, and statistical analysis.
- Currently exploring medical foundation models through concept-aware prompting and structured anatomical-text guidance, with an emphasis on prompt robustness, hard-negative concepts, and cross-dataset generalization.

PUBLICATIONS (* indicates equal contribution)

Accepted

- **Hekai Wang**, Xingyang Wu, Isah Salim Ahmad, Shuo Zhou, Zhao Liao, Caijin Zhou, Yihang Zhou, Dong Liang, Zhanqi Hu, and Haifeng Wang. AP-Mamba: Anatomy-Aware State Space Models for Multi-organ Abdominal MRI Segmentation. International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2026). Accepted; Oral Presentation.
- **Hekai Wang**, Zhenrong He, Xingyang Wu, Yihang Zhou, Dong Liang, and Haifeng Wang. Prior-Guided Flow Matching for Missing Modality MRI Synthesis. IEEE International Conference on Bioinformatics and Biomedicine (BIBM 2026). Accepted; Oral Presentation.
- Zhenrong He, Bin Zhang, **Hekai Wang**, Shuo Zhou, Xingyang Wu, Lixian Zou, Yihang Zhou, Haifeng Wang. ATLANTIS: An Anatomy-guided Tissue-Latent Atlas Network for DWI-derived White-Matter Anisotropy Synthesis from T1w MRI. IEEE International Conference on Biomedical and Health Informatics (BHI). Accepted; Oral Presentation.
- Isah Salim Ahmad, Tian Yu, Qiang Zhang, Xingyang Wu, Zhao Liao, **Hekai Wang**, Xia Zhao, Bixia Yuan, Rongbo Lin, Cailei Zhao, Jianxiang Liao, Benqing Wu, Haifeng Wang, and Zhanqi Hu. Brain Microstructural Alterations in Children with Tuberous Sclerosis Complex: A Neurite Orientation Dispersion and Density Imaging Study. Brain Sciences.

Under Review

- **Hekai Wang**, Zhenrong He, Haobo Wang, Jinxuan Lv, Yanjie Zhu, Yihang Zhou, Dong Liang, Zhanqi Hu, and Haifeng Wang. ReLiSS: Modality-Adaptive Representation Conditioning for Multimodal MRI Segmentation under Arbitrary Modality Availability. IEEE Transactions on Medical Imaging (TMI). (***Under review***)
- **Hekai Wang**, Zhenrong He, Shengran Fan, Qiang Zhang, Yaxin Cai, Shuyuan Chen, Caijin Zhou, Xingyang Wu, Shuo Zhou, Xin Liu, Yihang Zhou, Zhanqi Hu, and Haifeng Wang. OA-CXR: An Anatomical Retention-Guided System for Selective Chest Radiograph Report Review. Expert Systems with Applications (ESWA). (***Under review***)
- **Hekai Wang**, Zhenrong He, Shengran Fan, Xingyang Wu, Caijin Zhou, Yihang Zhou, Dong Liang, Zhanqi Hu, and Haifeng Wang. Geometry-Aware Biplanar Chest CT Reconstruction with Residual Latent Diffusion. IEEE International Symposium on Biomedical Imaging (ISBI 2027). (***Under review***)

Manuscripts in Preparation

- **Hekai Wang**, et al. AnatoTextSeg: Multi-Organ Segmentation Guided by Structured Anatomical Text.

Patents

- Hu, Z., **Wang, H.**, Dai, J., Liu, R., Yu, T., Lin, L., Wu, X., Zhou, C., Zhao, Z., Wang, H., Wu, B., 2026. A Cerebrospinal Fluid Suppression-Constrained Magnetic Resonance Image Fusion Method and Related Apparatus. Chinese Invention Patent, Application No. 202611682545.4.

CLINICAL AND INDUSTRY EXPERIENCE

Shenzhen Guangming District People's Hospital — **Shenzhen, China**
Clinical Data and Predictive Modeling — **Dec 2025 – Present**

- Built structured clinical research databases and machine-learning pipelines for cholecystitis prediction, TSC diagnosis, and rehabilitation-outcome prediction, from data curation through model evaluation and clinical interpretation.

Shanghai United Imaging Healthcare Co., Ltd. — **Shanghai, China**
MRI Sequence Development and Reconstruction — **Dec 2024 – Feb 2025**

- Investigated MRI pulse-sequence development and reconstruction workflows, including ADEPT-based development, SMS/parallel imaging, GRE/FLAIR implementation, and Gadgetron-based inline reconstruction.

SKILLS & INTERESTS

Research: Medical Image Analysis, MRI Reconstruction and Synthesis, Multimodal Medical Imaging, Medical Image Segmentation, Generative Models.
Programming: Python, PyTorch, MATLAB, C++.
Technical: Medical image preprocessing (NIfTI, resampling, normalization), 2D/3D deep learning, Large Language Model, diffusion models, flow matching, distributed training.
Experimental: Cross-validation, ablation studies, multi-dataset evaluation, external validation, bootstrap confidence intervals, statistical significance testing.

ACADEMIC REFERENCES

Prof. Haifeng Wang
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Email: hf.wang1@siat.ac.cn
Relationship: Master's Research Supervisor

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Relationship: Master's Research Co-supervisor